

# The Historical Income Panel for the Netherlands

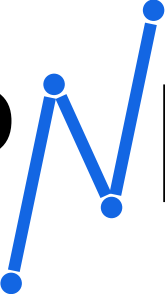
Auke Rijpma

February 2, 2026

- ▶ Economic and social historian at Utrecht University.
- ▶ Interested in reconstructing how past economies and societies worked, and bringing social scientific theories and methods.
- ▶ In particular interested in the long turn development of welfare & its drivers.
- ▶ Veteran of Clariah, Clariah Plus projects, longstanding interest in developing the infra needed to do the above.
- ▶ PI of the HIPNL project,

HIPNL

HIPNL

A stylized graphic of the letters "HIPNL". The letters "H", "I", "P", and "L" are in a bold, black, sans-serif font. The letter "N" is represented by a blue line connecting four blue dots. The dots are positioned at the top-left, top-right, bottom-left, and bottom-right of the "N" shape. The line starts at the bottom-left dot, goes up to the top-left dot, then down to the bottom-right dot, and finally up to the top-right dot.

# Today

- ▶ The Historical Income Project for the Netherlands, 1850-1920 (HIPNL).
  - ▶ Motivation
  - ▶ Fit in Dutch historical infra
  - ▶ Current state of project
  - ▶ Methods, and role of AI
  - ▶ Future plans

# Motivation

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W. Salverda and A. B. Atkinson

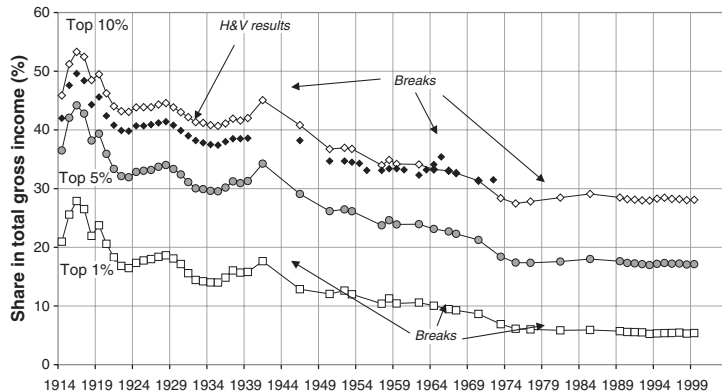


Figure 10.3A Gross income shares of top 10%, 5%, and 1%, Netherlands 1914–99

Figure 1: Income share in Netherlands

# Motivation

- ▶ HIPNL strives to address:
  - ▶ Inequality gap: despite renewed interest in inequality, period prior to WWI largely unknown.
  - ▶ Economic history gap: despite importance of microdata in econhis research, we lack *economic* microdata (unlike situation in social and demographic history).
  - ▶ Historical social graph/grail quest: it is possible to reconstruct the demographic, social, and economic lives and connections of all Dutch citizens going back to at least 1812, but we have not done so yet.

# Motivation

- ▶ HIPNL contributes to this by:
  1. collecting 100.000s of household-level income estimates for the period 1850–1920 on the basis of the *Hoofdelijke Omslag* taxes.
  2. Linking these households to the Dutch population registers (*Bevolkingsregisters*), which allow us to link to civil registry as a central linking hub in the Dutch historical microdata infrastructure.

## HIP-NL sources: Zaandijk 1879

[illegible]

# HIP-NL sources: Utrecht 1919

| Volg. nummer. | Naam, beroep en woonplaats der belastingsschuldigen                                              | Burg. staat en aantal kinderen. | Werkelijk inkomen. | Aftrek levensonderhoud. | Belastbaar inkomen. | AANSLAG.        |         |
|---------------|--------------------------------------------------------------------------------------------------|---------------------------------|--------------------|-------------------------|---------------------|-----------------|---------|
|               |                                                                                                  |                                 |                    |                         |                     | Aantal maanden. | Bedrag. |
| 3511          | Mej. G. v. Leeuwen Vert.:<br>M. v. Leeuwen, forstelijk<br>Agnesthuys, N° 2                       | 0/-                             | 950                | 500                     | 450                 |                 | 25 22   |
| 2             | Wed. H. M. Weijzenberg geb. W/-<br>T. Gardet Amsterdam, Florastr. 29.<br>N° 2                    | 1750                            | 550                | 1200                    |                     |                 | 77 41   |
| 3             | G. F. W. Bantz Cur: G. Bantz<br>Goheveningen, Haringkade 113.<br>N° 2 <i>Geo: naam Antwerpen</i> | 0/-                             | 800                | 500                     | 300                 |                 | 15 25   |
| 4             | C. Ripping Cur: Jhr. C. T.<br>Goldman te Eindhoven<br>N° 2 <i>Antlaag, Arnhemmed. N° 80</i>      | 0/-                             | 1150               | 500                     | 650                 |                 | 38 09   |
| 5             | K. M. Bikkema Veider Vert: G/-<br>J. J. Wieringa, huisgezand<br>N° 2                             | 0/-                             | 950                | 500                     | 450                 |                 | 25 22   |
| 6             | D. J. A. Tol Cur: Mevr. D. J. A.<br>A. Tol geb. A. Zimmerman<br>Franeker, Zuideweg 90.<br>N° 2   | 14/3                            | 2100               | 700                     | 1400                |                 | 92 23   |
| 7             | J. v. d. Wijk Prov. bevelind, G/-<br>H. G. v. d. Wiik te Hilversum<br>N° 2                       | 0/-                             | 2950               | 500                     | 2450                |                 | 172 96  |
| 8             | J. J. Wilmenga Vert: J. J. A.<br>Wilmenga te Garijp<br>N° 2                                      | 0/-                             | 900                | 500                     | 400                 |                 | 22 16   |
| 9             | A. Erainsheerd<br>Klerk 3. 3.<br>Amsterdamstr. N° 4bis                                           | 0/-                             | 650                | 500                     | 150                 |                 | 2 66    |
| 3520          | Mej. E. A. Huijgens<br>Handw. Oudewijzeres                                                       | 0/-                             | 700                | 500                     | 200                 |                 | 10 59   |

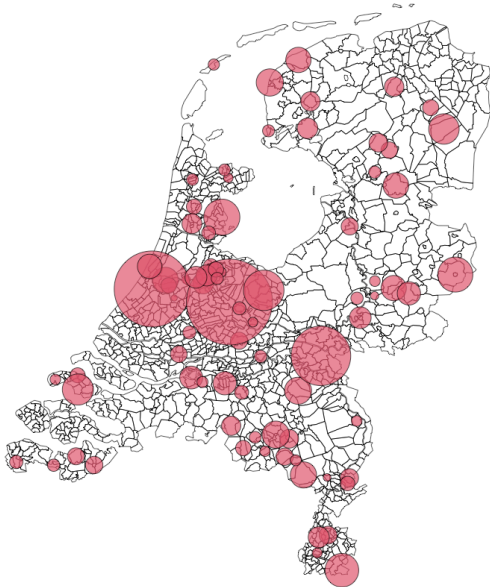
# HIP-NL sources: Leiden 1889

|                                            | 7  | Klasse. | Inkomen. | 1889. | 1888. | Belasting. |
|--------------------------------------------|----|---------|----------|-------|-------|------------|
| 111 J. P. A. Mulert . . . . .              | 8  | 1375    | 25,35    |       |       |            |
| 113 G. J. Bos . . . . .                    |    | 4000    | 93,60    | 87,10 |       |            |
| 115 J. Brevée . . . . .                    | 13 | 3250    | 74,10    | 74,10 |       |            |
| 115 T. van Veen . . . . .                  | 3  | 550     | 3,90     | 3,90  |       |            |
| 117 J. van der Staaij . . . . .            | 1  | 425     | 0,65     | 0,65  |       |            |
| <b>Nonnensteeg.</b>                        |    |         |          |       |       |            |
| 1 W. F. R. Suringar . . . . .              | 23 | 15000   | 372,—    | 372,— |       |            |
| 5 F. J. Struikenkamp . . . . .             | 2  | 475     | 1,79     | 1,79  |       |            |
| 7 H. Witte . . . . .                       | 12 | 2750    | 61,10    | 61,10 |       |            |
| 7 H. W. Witte . . . . .                    | 5  | 775     | 9,75     | 10,40 |       |            |
| 7 E. T. Witte . . . . .                    | 3  | 550     | 3,90     | 2,60  |       |            |
| 2 A. Teske . . . . .                       | 1  | 425     | 0,65     | 0,65  |       |            |
| 16 P. Brieër . . . . .                     | 1  | 425     | 0,65     | 0,65  |       |            |
| 18 F. C. J. Schnabel . . . . .             | 1  | 425     | 0,65     | 0,65  |       |            |
| 18 J. T. Schnabel . . . . .                | 1  | 425     | 0,65     | 0,65  |       |            |
| 20 J. Evers . . . . .                      |    | 600     | 4,88     | 4,88  |       |            |
| <b>Kolfmakerssteeg.</b>                    |    |         |          |       |       |            |
| 4 A. van der Vecht . . . . .               | 4  | 650     | 6,50     | 6,50  |       |            |
| 7 A. Versnij . . . . .                     | 2  | 475     | 1,87     | 1,84  |       |            |
| 14 W. van Ingen . . . . .                  | 1  | 425     | 0,59     | 0,59  |       |            |
| 24 C. T. H. Mook . . . . .                 | 3  | 550     | 3,58     | 0,62  |       |            |
| <b>Zegersteeg.</b>                         |    |         |          |       |       |            |
| 1 G. van der Reijden . . . . .             | 1  | 425     | 0,65     | 0,65  |       |            |
| <b>Doelenachtergracht.</b>                 |    |         |          |       |       |            |
| 2 H. Frankhuijzen . . . . .                | 1  | 425     | 0,65     | 0,65  |       |            |
| 19 J. Ober . . . . .                       | 1  | 425     | 0,62     | 0,62  |       |            |
| <b>Kaiserstraat.</b>                       |    |         |          |       |       |            |
| 1 J. van der Poel . . . . .                | 6  | 925     | 12,29    | 12,29 |       |            |
| 1 P. B. van Oerle . . . . .                | 3  | 550     | 3,90     | 3,90  |       |            |
| 3 J. F. Neuteboom . . . . .                | 1  | 425     | 0,65     |       |       |            |
| 3 C. H. Beij . . . . .                     | 2  | 475     | 1,95     |       |       |            |
| 5 W. F. K. Buitenweg . . . . .             | 1  | 425     | 0,65     | 0,65  |       |            |
| 5 J. van Ulden . . . . .                   | 1  | 425     | 0,65     | 0,62  |       |            |
| 7 W. Epskamp . . . . .                     | 5  | 775     | 8,58     | 9,75  |       |            |
| 7 J. N. van der Mark . . . . .             |    | 900     | 12,22    | 12,22 |       |            |
| 9 H. L. van Nifterik . . . . .             | 12 | 2750    | 59,87    | 61,10 |       |            |
| 11 Wed. J. v. d. Meulen-Wakerlin . . . . . | 1  | 425     | 0,62     | 0,62  |       |            |
| 13 D. Jurriaanse . . . . .                 | 8  | 1375    | 25,35    | 25,35 |       |            |
| 13 J. Groot . . . . .                      | 7  | 1125    | 18,85    | 18,85 |       |            |
| 19 C. Collee . . . . .                     | 3  | 550     | 3,58     | 3,66  |       |            |
| 27 D. Segeyn . . . . .                     | 1  | 425     | 0,62     | 0,62  |       |            |
| 33 L. de Leeuw . . . . .                   | 1  | 425     | 0,62     | 0,62  |       |            |
| 39 T. N. Schotman . . . . .                | 3  | 550     | 3,90     |       |       |            |
| 47 H. H. van Barmburgh . . . . .           |    | 4000    | 44,97    | 48,09 |       |            |

## In the larger infrastructure.

- ▶ HSN: the original large historical microdata project. HIPNL adds income estimates to 10% of HSN's observations.
- ▶ LINKS: all births, marriages, and deaths. All linked observations in HIPNL will
- ▶ CLARIAH: Linked Data Workbench for SSH + LD instance of LINKS, to which HIPNL can be added.
- ▶ Lifetree-NL: ambition to add HSN, LINKS, HIPNL, etc. to present-day SSD.
- ▶ Archival scans and indexing projects: HIPNL would be impossible without investments by archives to digitise and index their materials.

## HIP-NL sample



## Current state

- ▶ Sample of municipalities, stratified by province, municipality size, aiming to cover 10% of population in 7 sample years.
- ▶ Resulting database has income and tax data for 350k households, from 78 municipalities.
- ▶ Linked to population and civil registries for c. 50% of data.

# Methods

- ▶ At first, classic #econhis project:
  - ▶ Lots of RAs scanning and transcribing scans.
  - ▶ Traditional OCR for the printed volumes.
  - ▶ ML (boosting) for the linkage.
- ▶ HIPNL coincided with the LLM revolution, and we radically changed our approach.

# Methods

- ▶ OCR: traditional issues: character error rate good enough, but post-processing very time intensive.
- ▶ NOCR:
  - ▶ DONUT: finetune our own pre-VLM to directly create structured data (promising but still time-intensive).
  - ▶ In the end, leverage commercial VLMs (Gemini) in a few-shot learning approach.

| 187         |                                              |                                                   |         |                                             |       | W I J K                     |    |    |  |  |  |
|-------------|----------------------------------------------|---------------------------------------------------|---------|---------------------------------------------|-------|-----------------------------|----|----|--|--|--|
| Volgnummer. | N A M E N<br>VAN DE<br>BELASTINGSSCHULDIGEN. | WOONPLAATS.<br>Straat, gracht, enz.<br>en nummer. | Klasse. | Bedrag<br>van<br>den aanslag<br>à<br>3 pet. |       | N <sup>o</sup> .<br>van het |    |    |  |  |  |
|             |                                              |                                                   |         |                                             |       |                             |    |    |  |  |  |
| 11061.      | G. Fukkink                                   | Oudekamp<br>N <sup>o</sup> . 10                   | 2       | 1                                           | 4 50  | 111                         | 14 | 59 |  |  |  |
| 11062.      | C.B. Beijerman                               | N <sup>o</sup> . 12                               |         | 10                                          | 60    | 53                          |    |    |  |  |  |
| 11063.      | J. van Hijtveld                              | N <sup>o</sup> . 16                               | 2       |                                             | 4 50  | 15                          |    |    |  |  |  |
| 11064.      | J.C.E. Procksie                              | N <sup>o</sup> . 16                               | 1       |                                             | 1 50  | 74                          |    |    |  |  |  |
| 11065.      | G. van Baak                                  | N <sup>o</sup> . 10                               | 9       |                                             | 49 50 | 8                           |    |    |  |  |  |
| 11066.      | J.H. Kneebout                                | Oudekerkhof<br>N <sup>o</sup> . 3                 | 1       |                                             | 1 50  | 45                          |    |    |  |  |  |
| 11067.      | J. van der Pijl                              |                                                   | 2       |                                             | 4 50  | 112                         |    |    |  |  |  |

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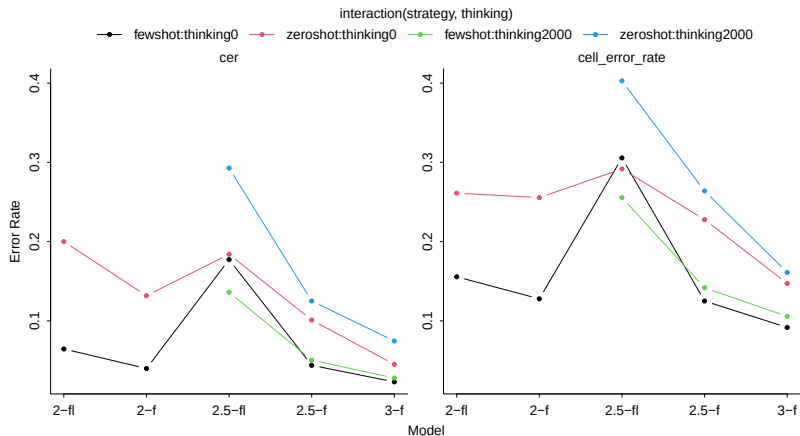
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}

# NOCR: error rates

Error Rates by Model, Strategy, and Thinking Budget



# NOCR

- ▶ Able to circumvent nearly all postprocessing, left with only more straightforward character corrections.
- ▶ Low cost (for us: free; now: 1000s of scans for c. \$1)
- ▶ Cornerstones:
  - ▶ Strong image comprehension of Gemini models
  - ▶ Strong schema adherence and prompt mapping.
  - ▶ In-context consistency: identifier perfectly remembered (2bdf4d0c-7ce1-40f6-aceb-0b85a2768881)

## Further LLM use

- ▶ Categorise (open weights, facebook/huggingface) names as male/female.
- ▶ Link checks:
  - ▶ Built an ML-based (gradient boosting) approach
  - ▶ Verified links with LLMs combined with human review.
  - ▶ Good at dealing with missing data, determining ambiguous link candidates.

# AI reflections

- ▶ While using AI tools boosts productivity, there are drawbacks.
  - ▶ Commercial closed v. commercial open v. open source
  - ▶ Reproducibility
  - ▶ Costs
  - ▶ Privacy
- ▶ Good scans and open data (potential training/finetuning) key to enable these pipelines.

## Future plans

- ▶ HIPNL new ambitions
- ▶ Ingesting newly available data such as population registers, tax records that keep coming online.
- ▶ Finish connections to other historical microdata to make HIPNL a broader “welfare” panel (health, wealth, education).